

Lab Sheet 1 (CST-108)

Unix Signals

1. Write a C program that installs a handler for SIGINT signal. Print some message inside signal handler. Write an infinite for loop at the end of the main function. Now compile and run this program. Press CTRL-C to terminate this program. What happens? Press CTRL-C again. What happens? Does your OS support reliable / unreliable Unix signals?
2. Write a C program that forks 5 children. When child number i terminates, it returns the exit status as i to the parent. Parent installs a signal handler for SIGCHLD signal. In signal handler parent collects and prints the exit status of each child along with the process id of the child. Study the output.
3. Modify the code in 2 above so that each child i sets an alarm that expires after $i*10$ seconds. Child process terminates when the alarm expires. Parent prints the process id of the child and its exit status in the signal handler. Is the output same as in Q2?
4. Modify the code in 3 above so that before terminating the parent prints the number of seconds spent by the parent in the user mode and in the system mode. It also prints the time spent by its children in the user mode and in the system mode.

[Note: use "getrusage" function. Convert the time given in microseconds to seconds and add it into the time given in seconds]

5. Write a C program that blocks the delivery of ONLY SIGALRM and SIGINT signals (use function sigprocmask). It sets an alarm that expires after 10 sec and then waits for 5 min. Before terminating it prints the name **and number** of signals that are pending for delivery. Compile and run this program. During the wait time of 5 min press CTRL-C twice.